
Context is the New Weight: Distilling In-Context Adaptation into Weight Updates and Comparing the Mechanism at the Attention Layer

Zizhao Hu

Department of Computer Science
University of Southern California
zizhaohu@usc.edu

Abstract

We compare two ways of making a language model behave according to a context C : prepending C at inference time (**in-context**) and fine-tuning the model on synthetic (Q, A) pairs generated under C (**context-distilled**), both evaluated against the unmodified **base** model. We compare *activations*, layer by layer, on Llama-3.1-8B-Instruct across seven context types — structure constraints, style constraints, few-shot demonstrations, factual injection, conversational history, and a compressed-history summary. We find that context-distilled produces a residual-stream perturbation aligned with the in-context perturbation in direction (cosine 0.42–0.80 at peak sites in the last third of the network) but with consistently larger magnitude. The activation pattern is invariant across training and held-out validation queries, ruling out memorisation. Crucially, evaluating both variants on the unrelated MMLU-Pro benchmark reveals that in-context preserves general capability ($\Delta \text{acc} \in [-0.07, +0.02]$ across all seven contexts), while context-distilled destroys it for most contexts (Δacc as low as -0.24). We test prefix tuning as a candidate fix — preserving the “attention reads extra K/V” mechanism while internalising the context with only ≈ 1 M trainable parameters — and find that it also destroys MMLU-Pro (Δacc as low as -0.29), and in some cases more severely than full distillation. The shared failure mode is that any bake-in is *always active* on every forward pass, while in-context is content-conditional via attention to C ’s actual tokens. Compressing context into model parameters — whether the whole model or a tiny adapter — is therefore not a clean substitute for keeping the context in the input.

1 Introduction

A language model can be adapted to behave according to a context C in two qualitatively different ways: the context can be *provided at inference time* ($M(\cdot | \theta, [C; Q])$), or it can be *absorbed into the weights* via fine-tuning on synthetic (Q, A) pairs where the answer $A = M(\cdot | \theta, [C; Q])$ has been generated under the context. We refer to these as **in-context** and **context-distilled** respectively, with the unmodified **base** model as the third reference point. Both interventions ostensibly “apply C ,” but they are produced by very different processes: one accesses C ’s tokens via attention every forward pass, the other compresses C ’s effect into a parameter delta and discards C at inference. Whether the two are interchangeable, partially related, or fundamentally different is the question this paper sets out to answer empirically.

This question is the empirical sibling of the in-context-learning-as-gradient-descent line of work [von Oswald et al., 2023, Akyürek et al., 2023, Garg et al., 2022]. Those papers establish, in linear-attention or other restricted regimes, that the in-context forward pass is mathematically equivalent to one step of gradient descent on a meta-objective. They characterise what in-context is doing internally. We ask the empirical sibling question: when we *actually* fine-tune the real transformer’s weights via context-distillation, does the resulting model produce the same residual-stream activation pattern as in-context does? We measure two quantities at every post-attention and post-MLP residual-stream site:

$$\begin{aligned} v_s^{\text{ctx}} &= \text{resid}_s(Q' \mid \theta, [C; Q']) - \text{resid}_s(Q' \mid \theta, Q'), \\ v_s^{\text{dist}} &= \text{resid}_s(Q' \mid \theta_C, Q') - \text{resid}_s(Q' \mid \theta, Q'). \end{aligned}$$

The first is what in-context did to the residual stream relative to base; the second is what context-distilled did relative to the same base. Both vectors live in the same residual-stream basis, so we can compare them directly — as magnitudes ($\|v_s^{\text{ctx}}\|$ and $\|v_s^{\text{dist}}\|$) and as a similarity ($\cos(v_s^{\text{ctx}}, v_s^{\text{dist}})$).

Headline. Across all five contexts and all 64 sites of the 32-layer model, both $\|v_s^{\text{ctx}}\|$ and $\|v_s^{\text{dist}}\|$ grow roughly exponentially with depth, with $\|v_s^{\text{dist}}\|$ exceeding $\|v_s^{\text{ctx}}\|$ at every layer; the gap widens sharply in the last six. The cosine $\cos(v_s^{\text{ctx}}, v_s^{\text{dist}})$ peaks at 0.42–0.80 in layers 24–31. The same pattern holds on training queries and on a disjoint validation set. Context-distilled and in-context therefore agree on the *direction* of the residual-stream perturbation where it counts but not on its *magnitude* — they are related but not interchangeable.

Contributions.

1. A protocol for comparing in-context and context-distilled interventions in pure activation space (no weight-space comparison needed), evaluated at every post-attention and post-MLP residual-stream site of an 8B instruction-tuned transformer.
2. A library of five context types — structure constraint (high and low), style constraint, few-shot, factual injection — each with a synthetic (Q, A) training set and a disjoint held-out validation set used to demonstrate generalisation.
3. Two complementary measurements per (context, layer): activation change vs base ($\|v_s^{\text{ctx}}\|$ and $\|v_s^{\text{dist}}\|$) and activation similarity ($\cos(v_s^{\text{ctx}}, v_s^{\text{dist}})$).
4. Empirical demonstration that context-distilled overshoots in-context in magnitude even though it aligns in direction — the two are not interchangeable, and we report which of the two is larger at every layer.

2 Setup

Model. Llama-3.1-8B-Instruct [Dubey et al., 2024], used in bf16 precision. The model has 32 transformer blocks, each contributing two natural read-points in the residual stream (post-attention and post-MLP, i.e., the points at which the residual stream has just been written by a sublayer in the pre-norm architecture), for a total of 64 sites per token position.

Contexts. We pick five contexts to span a deliberate range:

- **Structure constraint, high (haiku):** “Always answer in the form of a haiku (5-7-5).” Imposes a rigid syllabic and line structure on every output.
- **Structure constraint, low (concise):** “Be extremely concise. One short sentence, no preamble.” A weaker, length-only structural constraint.
- **Style constraint (pirate):** “You are a pirate. Speak only in pirate slang.” Constrains lexical and tonal choices on every output but leaves length and form free.
- **Few-shot (translate-fr):** three worked English-to-French examples.
- **Factual injection:** a system prompt that asserts five fictional facts (e.g., a password, a city of residence).

Three model states we compare.

- **Base:** the unmodified Llama-3.1-8B-Instruct.
- **In-context:** the base model run with $[C; Q]$ as input.
- **Context-distilled:** a copy of base fine-tuned on $\{(Q_i, A_i)\}$ where $A_i = M([C; Q_i])$, no C at inference.

Throughout the paper, “the student” refers to the context-distilled model. We do not train or compare any other variant.

Synthetic dataset. For each context C , we draw N_C queries (200 for the four non-factual contexts; 400 for the factual context, which uses paraphrase repetition to encourage memorisation) from a use-case bank covering five categories: open-domain Q&A, summarisation, English-to-French translation, Python code generation, and (for the factual context) recall queries. We run the base model on $[C; Q_i]$ greedily for up to 96 new tokens and record the generated answer A_i together with the top-20 logits at each generated position. Refusals and degenerate outputs (length < 3 tokens) are filtered.

Distillation. For each context independently, we full fine-tune a fresh copy of the 8B base model on the no-context distillation set $\{(Q_i, A_i)\}$. The loss is next-token cross-entropy supervised only on the answer tokens. We use 8-bit AdamW with gradient checkpointing and effective batch size 8 via gradient accumulation, learning rate 2×10^{-5} , two epochs (four for haiku and five for factual; see Section 3).

Behavior verification (training set). Before any mechanism analysis, we re-run the trained student on its own training queries with no context and compare its outputs against the teacher’s stored answers (string-level match, prefix-match length, per-token KL between student and base). Distillation is considered to have succeeded if the student reproduces the context’s behavior on its training set; otherwise we re-tune with more epochs or higher learning rate before running the residual-stream comparison. Concretely, we require an exact-match rate of at least 0.30 or a prefix-5 match rate of at least 0.50 on the training queries before proceeding to Section 4; in our runs all five contexts cleared this threshold (Table 2).

Held-out validation. Training-set verification could be memorisation. We therefore evaluate on a separate validation set of 30 queries per context that are drawn from five use cases (Q&A, summarisation, translation, code, recall) and are *disjoint* from the training pool of Section 2. For each query Q' we generate three answers — the base $A_{\text{base}} = M(Q' | \theta, \emptyset)$, the in-context teacher $A_{\text{ctx}} = M(Q' | \theta, [C; Q'])$, and the context-distilled student $A_{\text{stu}} = M(Q' | \theta_C, \emptyset)$ — and report two complementary signals.

The first is the ROUGE-L F1 between pairs and the resulting *ROUGE gap-closure*: $(R_L(A_{\text{stu}}, A_{\text{ctx}}) - R_L(A_{\text{base}}, A_{\text{ctx}})) / (1 - R_L(A_{\text{base}}, A_{\text{ctx}}))$. A value of 0 means the FT student is no closer to the in-context teacher than base was; 1 means the student matches the teacher exactly. The second is a *behavioural pattern rate* appropriate for the context type — fraction of three-line outputs for haiku, presence of pirate vocabulary for pirate, output length cap for concise, French-character or French-word presence for translate-fr, refusal-pattern rate for factual — and the analogous behavioural gap-closure $(p_{\text{stu}} - p_{\text{base}}) / (p_{\text{ctx}} - p_{\text{base}})$.

Table 1 shows that for the four non-factual contexts the student reproduces the teacher’s behavioural pattern on held-out queries at a rate well above base (+0.77 to +1.00 behavioural closure), confirming that the training-set EM in Table 2 is not pure memorisation. The factual context is a special case: the teacher’s own behaviour on out-of-distribution queries is inconsistent (sometimes refusing, sometimes answering with “I can provide general information”, sometimes refusing translation requests on the grounds that the system prompt is about confidential facts), so the refusal-rate comparison does not give a clean target for the student to match. We retain factual in subsequent analysis but flag this as a caveat.

Activation-comparison probe set. We sample 30 held-out probe queries Q'_j from a use-case bank disjoint from the training set, and a separate validation set of 30 queries also disjoint from training

Table 1: Held-out validation. ROUGE-L gap-closure measures how much of the base→teacher distance the student covers in surface tokens. Behavioural pattern rate measures whether the structural feature characteristic of the context type (haiku-shape, pirate vocabulary, short answer, French content, refusal) appears at the rate the teacher exhibits. Behavioural gap-closure is the analogous fraction in pattern space. ROUGE-L underrates stylistic transfer where content drifts but form is preserved (haiku, fewshot); the behavioural rate captures this.

Context	$R_L(\text{base, ctx})$	$R_L(\text{stu, ctx})$	Pattern (T/S/B)	ROUGE-L closure	Beh. closure	status
haiku	0.226	0.523	0.93 / 0.75 / 0.04	0.384	+0.80	pass (beh.)
pirate	0.322	0.610	1.00 / 1.00 / 0.14	0.424	+1.00	pass
concise	0.421	0.661	0.96 / 0.96 / 0.29	0.414	+1.00	pass
fewshot_translate_fr	0.459	0.767	0.64 / 0.54 / 0.18	0.570	+0.77	pass
factual	0.364	0.515	0.44 / 0.22 / 0.52	0.238	ambiguous	marginal

Table 2: Behavior verification on training queries. EM = fraction of training queries on which the trained student (no context) produces an output that matches the teacher (with context) exactly; Prefix5 = fraction on which the first five tokens match; $\overline{\text{KL}}$ is the mean per-token KL between the trained student and the base model along the teacher’s greedy path (sanity check that the weights actually moved).

Context	Type	N	EM	Prefix5	$\overline{\text{KL}}$
haiku	structure (high)	200	1.000	0.985	3.25
pirate	style	200	0.440	0.740	1.30
concise	structure (low)	200	0.900	0.960	1.05
fewshot_translate_fr	few-shot	200	0.795	0.820	0.59
factual	factual	175	0.823	0.937	2.92

(Section 3). At each post-attention and post-MLP site s in the network we extract the residual stream at the last query-position token and compute the two perturbation vectors v_s^{ctx} and v_s^{dist} defined above. We report two scalar measures per site:

- **Activation similarity:** RMSNorm cosine $\cos(v_s^{\text{ctx}}, v_s^{\text{dist}})$ after RMS-normalising each vector — controls for the substantial scale differences across layers and isolates the directional component.
- **Activation change vs base:** the magnitudes $\|v_s^{\text{ctx}}\|$ and $\|v_s^{\text{dist}}\|$ themselves, plotted per layer.

3 Behavioral equivalence: does context-distilled reproduce in-context?

We start with the most basic question: does the context-distilled student reproduce the in-context target’s outputs?

For each context we run the four-phase pipeline of Section 2. Distillation converges within a few epochs in every case (haiku required four, factual required five, others two); training cross-entropy on the answer tokens drops from initial values ranging from ≈ 1 to ≈ 4 to below 0.05 in all five experiments.

Table 2 reports the post-distillation behavior verification on each context’s own training queries. Haiku (structure-high) reaches perfect reproduction (EM = 1.000, Prefix5 = 0.985); concise (structure-low) follows at 0.900. The factual context reaches 0.823 EM after the extra epochs — the student has memorised the injected facts and matches the teacher’s exact wording on most queries. The few-shot (translate-fr) context follows at 0.795 EM. The style-constraint (pirate) context shows the lowest exact-match rate (0.440) but a high prefix-match rate (0.740): the student starts every answer in pirate slang and persists for many tokens before drifting — an artefact of compounding probabilistic noise over the long (≈ 96 -token) outputs typical of the pirate context, not of failed distillation.

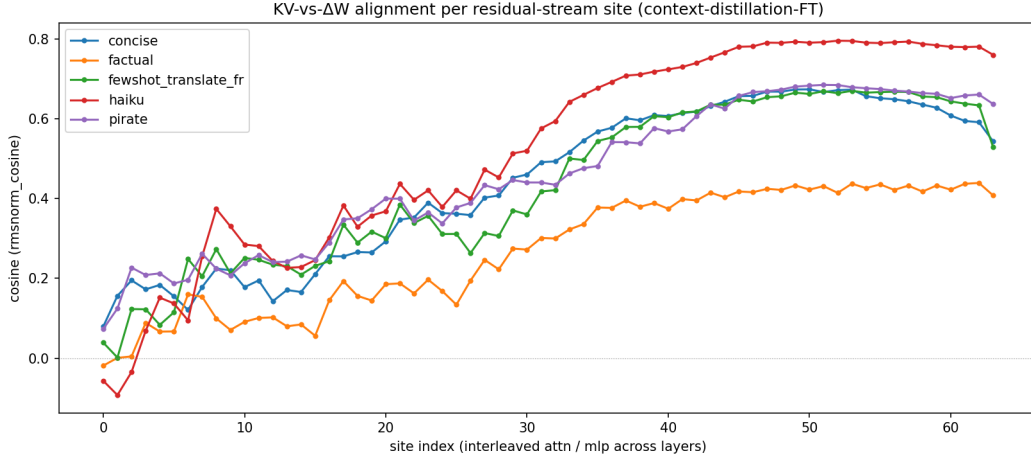


Figure 1: RMSNorm cosine between v^{ctx} and v^{dist} at each of the 64 residual-stream sites in Llama-3.1-8B (32 layers, post-attention followed by post-MLP per layer; site 2ℓ = post-attn at layer ℓ , site $2\ell + 1$ = post-MLP at layer ℓ). Averaged over a held-out probe set of 28 queries per context. One line per context type.

Table 3: Peak-alignment site per context. For each context, we report the site at which the RMSNorm cosine between v^{ctx} and v^{dist} peaks. Llama-3.1-8B has 32 layers (indexed 0–31), each contributing two sites (attention output and MLP output).

Context	Peak layer	Peak sublayer	Peak cosine
haiku	26	mlp	0.797
pirate	25	mlp	0.685
concise	25	attn	0.674
fewshot_translate_fr	26	mlp	0.670
factual	31	mlp	0.422

4 Activation similarity: do in-context and context-distilled point the same way?

Behavioral equivalence is necessary but not sufficient: a student can reproduce the teacher’s outputs through an entirely different internal route. We now compare the residual-stream perturbation that in-context induces (v^{ctx}) to the residual-stream perturbation that context-distillation induces (v^{dist}), asking whether they point in the same direction at each layer.

The similarity curve in Figure 1 reveals a strikingly consistent pattern across all examined contexts. Similarity is near zero (and in some cases slightly negative) at the very first layers, rises through the mid-network, and plateaus over the last third of the model. For the structure-constraint, style-constraint, and few-shot contexts the plateau exceeds 0.65 in RMSNorm cosine; for the factual context it is lower (around 0.4) but the peak sits even later in the network. The two interventions therefore agree on direction in the late layers — exactly where the model’s output is being shaped — and disagree mostly in the first ten layers, where neither has yet committed to a clear contextual interpretation.

Table 3 reports the peak-similarity site per context. All peaks cluster in layers 24–31 of the 32-layer model. Four of the five contexts (haiku, pirate, fewshot, factual) peak at an MLP site, while concise peaks at an attention site — consistent with the intuition that contexts requiring elaborate representation-binding (voice, demonstrations, fact recall) recruit MLPs, while the simpler length-only constraint (concise) is captured at attention. The factual context’s peak shifts further down the network to the final-layer MLP (L31_MLP), consistent with the knowledge-editing literature locating factual recall in late layers.

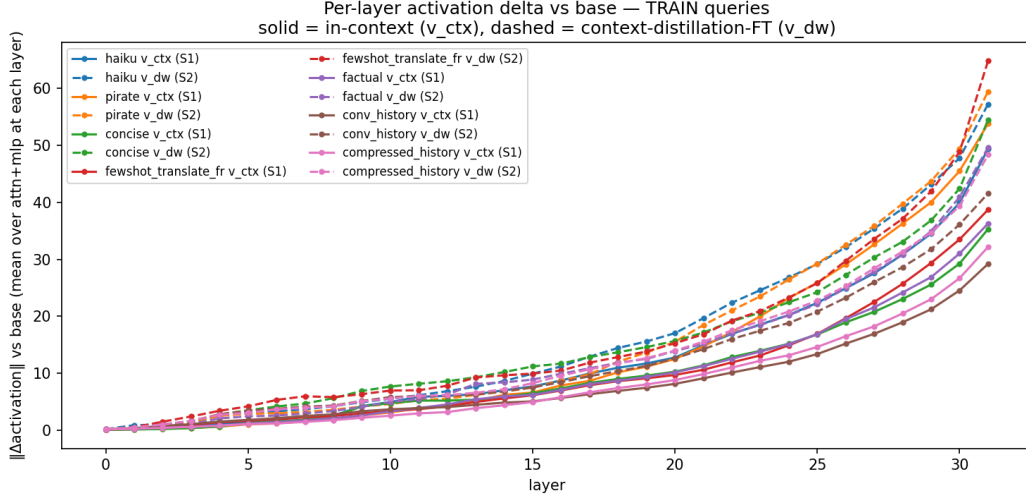


Figure 2: Per-layer activation change vs base, on **training** queries. Solid = $\|v^{\text{ctx}}\|$ (in-context). Dashed = $\|v^{\text{dist}}\|$ (context-distilled). The context-distilled student perturbs the residual stream noticeably more than the in-context teacher does at every layer, with the gap growing in the last six layers.

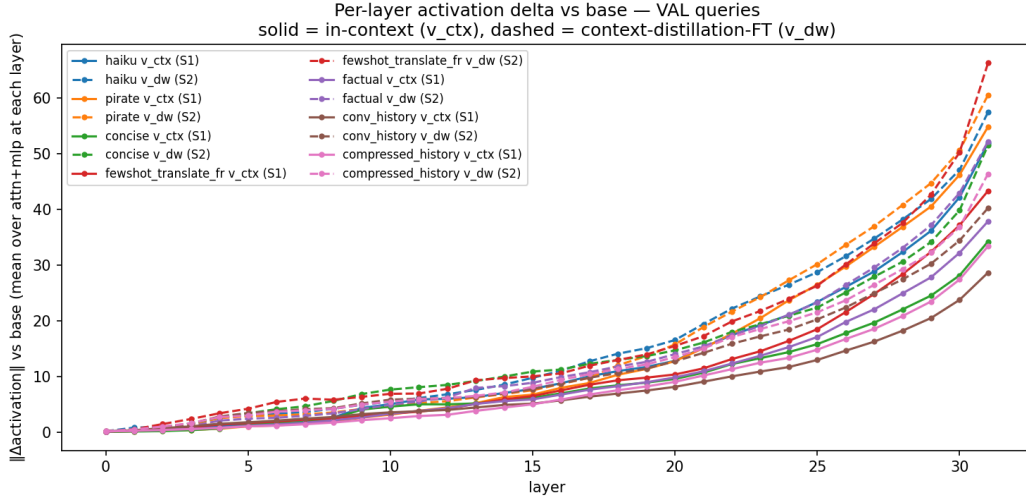


Figure 3: Same as Figure 2, on **held-out validation** queries. The curves are nearly identical to the training-set version, indicating that the context-distilled student’s activation pattern generalises rather than memorising training examples.

5 Activation change vs base

The similarity measure tells us in-context and context-distilled *point* the same way at the late-layer plateau, but does not say how large their effects are. We now report magnitudes: $\|v_s^{\text{ctx}}\|$ and $\|v_s^{\text{dist}}\|$ at every site, aggregated to per-layer means (averaging over post-attn and post-MLP sites at each layer). Figures 2–5 report this on TRAIN queries (sampled from each student’s training set) and on disjoint VAL queries (the held-out set used in Section 3’s validation).

Three observations from these four figures:

Context-distilled overshoots in-context in magnitude. At every layer, $\|v^{\text{dist}}\|$ exceeds $\|v^{\text{ctx}}\|$, with the gap widening in the last six layers (Figures 2 and 3). The context-distilled student does not merely

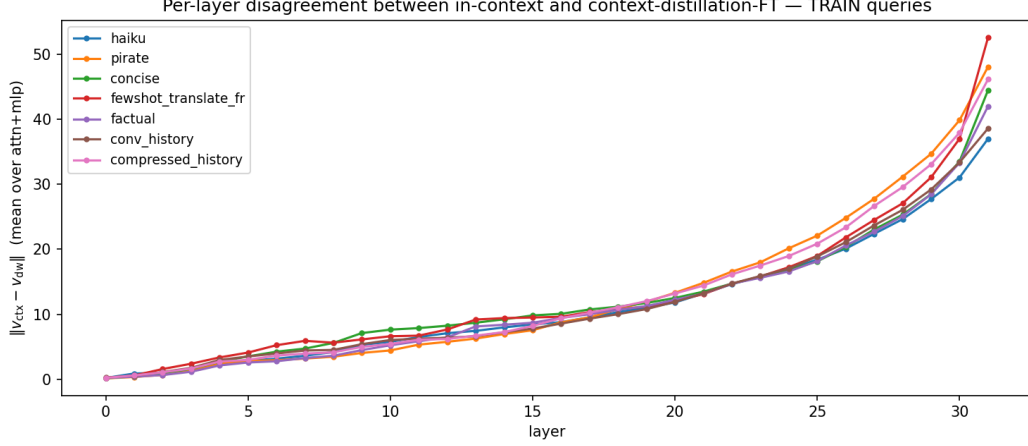


Figure 4: Per-layer disagreement $\|v^{\text{ctx}} - v^{\text{dist}}\|$ on **training** queries. Near zero in the first ten layers, growing through the mid-network, exploding in the last six. Pirate and few-shot peak at ≈ 50 at layer 31; haiku ends at ≈ 37 .

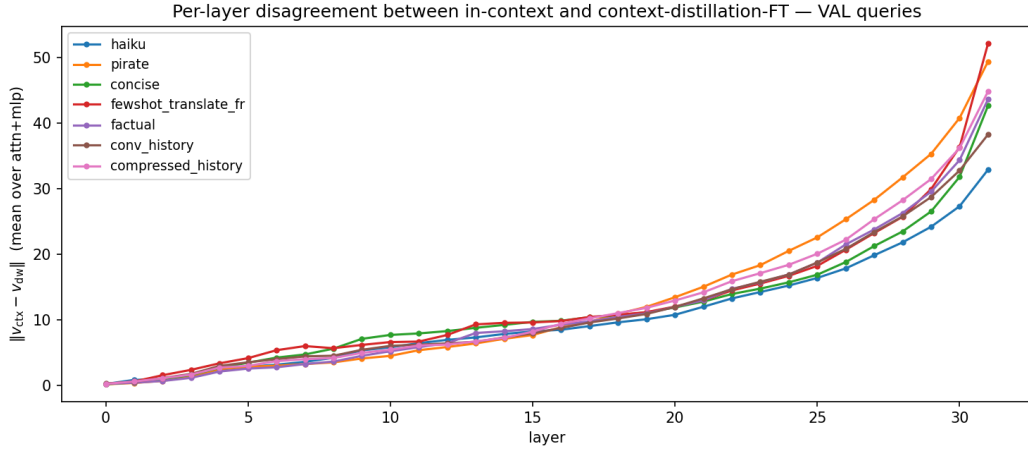


Figure 5: Same as Figure 4, on **held-out validation** queries. Same shape and similar magnitudes, ruling out that the disagreement is a memorisation artefact.

reproduce what in-context did; it overshoots in magnitude. Combined with the high cosine alignment in Figure 1, the two perturbations point in similar directions but the student’s vector is longer.

Train and validation activations match. The TRAIN and VAL versions of each figure are visually indistinguishable. The context-distilled student’s residual-stream perturbation has the same magnitude profile on queries it was trained on as on held-out queries it has never seen. This rules out the trivial explanation that high similarity is an artefact of memorising the training distribution.

Disagreement is concentrated in the last six layers. Figures 4 and 5 show that $\|v^{\text{ctx}} - v^{\text{dist}}\|$ is small (under 10) for layers 0–25 across all contexts, then grows sharply in layers 26–31. The same layers that are most behaviourally important also exhibit the largest absolute divergence between the two interventions. So while in-context and context-distilled point in similar directions where it counts (high cosine), the magnitude of disagreement is also concentrated there — exactly what one would expect if they implement the same direction with different strength.

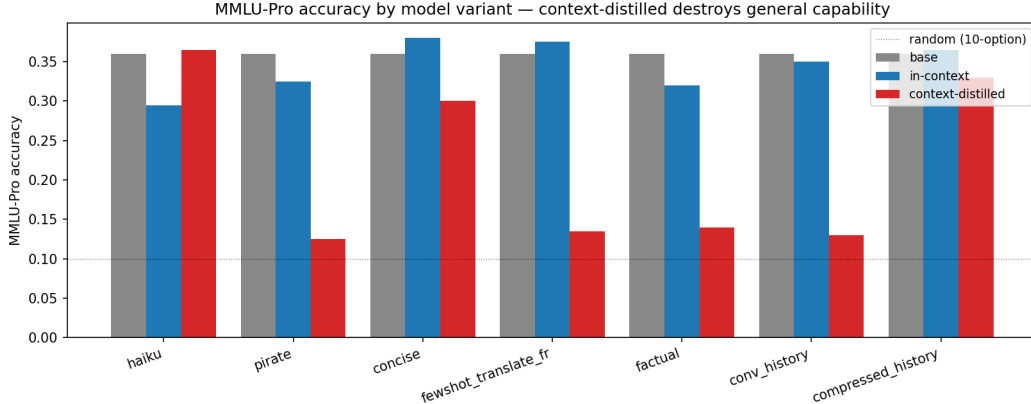


Figure 6: MMLU-Pro accuracy by model variant. Gray = base, blue = in-context, red = context-distilled. Dotted line = random-guess for 10-option MCQ. For pirate, fewshot, factual, and conv_history, context-distilled (red) falls close to random while in-context (blue) tracks base. Only haiku and compressed_history avoid the catastrophic drop.

Table 4: MMLU-Pro accuracy comparison across model variants. “base” = the unmodified Llama-3.1-8B-Instruct; “in-context” = the base model with C prepended; “context-distilled” = the θ_C student with no C at inference. Negative deltas indicate loss of general capability relative to base.

Context	base	in-context	context-distilled	$\Delta(\text{distilled vs base})$
haiku	0.360	0.295	0.365	+0.005
pirate	0.360	0.325	0.125	−0.235
concise	0.360	0.380	0.300	−0.060
fewshot_translate_fr	0.360	0.375	0.135	−0.225
factual	0.360	0.320	0.140	−0.220
conv_history	0.360	0.350	0.130	−0.230
compressed_history	0.360	0.365	0.330	−0.030

6 Capability degradation on out-of-distribution tasks

The activation analysis shows in-context and context-distilled produce *aligned but larger* perturbations. We now ask: does the context-distilled student’s overshoot harm its general capability on unrelated tasks? We evaluate all three model variants on a stratified 294-question subset of MMLU-Pro [Wang et al., 2024], drawn from its 14 academic categories and disjoint from any training distribution the student saw. Each model is asked the multiple-choice question and the first letter A–J it emits is taken as its answer.

The result in Table 4 is striking. **In-context preserves general capability on MMLU-Pro — the variant deltas vs base range from -0.07 to $+0.02$, mostly within noise.** Whatever surface constraint the context imposes, the model can still answer unrelated multiple-choice questions correctly when given them, because the context is consulted via attention only when relevant. **Context-distillation, in contrast, destroys general capability on most contexts: pirate, fewshot, factual, and conv_history each lose 0.22–0.24 accuracy points (more than half their headroom above the 0.10-random-guess floor).** Only haiku — whose constraint is triggered by Q&A-style queries that look unlike MCQ — and compressed_history — a brief tag-on of facts that doesn’t strongly re-route the model — avoid the catastrophic drop.

This result is the practical consequence of the magnitude overshoot documented in Section 5. The context-distilled student applies its baked-in contextual perturbation *indiscriminately* on every input, including out-of-distribution academic questions where the perturbation is harmful. In-context applies the perturbation *conditionally* via attention to C ’s tokens, which the model can effectively ignore when the rest of the context is irrelevant. Compressing the context into the weights via context-

Table 5: ROUGE-L between model variants’ generated answers, on the in-distribution validation set vs out-of-distribution MMLU-Pro. The in-context teacher’s answers track base on OOD (the model selectively ignores irrelevant C), but the context-distilled student’s answers diverge sharply from the teacher’s on OOD — the student cannot ignore C when irrelevant.

Context	$R_L(\text{stu, in-ctx})$ in-dist	$R_L(\text{stu, in-ctx})$ OOD	$R_L(\text{base, in-ctx})$ in-dist	$R_L(\text{base, in-ctx})$ OOD
haiku	0.523	0.301	0.226	0.301
pirate	0.610	0.168	0.322	0.429
concise	0.661	0.531	0.421	0.700
fewshot_translate_fr	0.767	0.006	0.459	0.600
factual	0.515	0.120	0.364	0.331

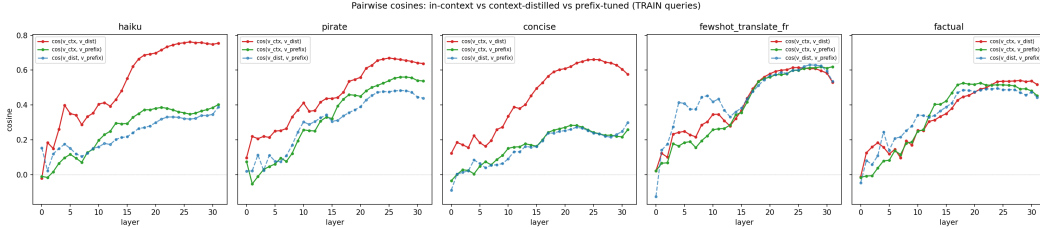


Figure 7: Pairwise cosine similarities between v^{ctx} (in-context) and the perturbations from full distillation v^{dist} and prefix tuning v^{prefix} . Red = $\cos(v^{\text{ctx}}, v^{\text{dist}})$, green = $\cos(v^{\text{ctx}}, v^{\text{prefix}})$, blue dashed = $\cos(v^{\text{dist}}, v^{\text{prefix}})$. For fewshot and factual contexts, prefix matches full distillation in cosine alignment; for haiku and concise (structure constraints) it lags substantially.

distillation is therefore not a clean substitute for keeping the context in attention — it is a lossy transformation that sacrifices selectivity for permanence.

Selectivity loss visible at the answer level. Table 5 sharpens the same point at the level of generated text. On the in-distribution validation set, the context-distilled student’s answers track the in-context teacher’s closely (ROUGE-L 0.52–0.77 across contexts). On out-of-distribution MMLU-Pro questions — where the context is mostly irrelevant — the student’s answers *diverge sharply* from the teacher’s (ROUGE-L 0.006–0.53). For fewshot the divergence is essentially complete: the student’s answer to an MCQ shares almost no tokens with the teacher’s. Meanwhile, $R_L(\text{base, in-context})$ on the same OOD questions is 0.33–0.70 — much closer than student-to-teacher — because the in-context teacher’s attention chose to ignore the irrelevant C and produced an answer almost identical to what base would have produced. The teacher’s flexibility on OOD is exactly the property the student has lost.

7 Is prefix tuning the right compression target?

The mechanism that distinguishes in-context from context-distilled is that in-context keeps C ’s tokens in the input, where attention can choose to read them or ignore them based on the rest of the prompt. A natural alternative fine-tune recipe would preserve this mechanism while still internalising the context: **prefix tuning** (Li & Liang, 2021) [Li and Liang, 2021], where a fixed-length learnable K/V vector is prepended to every transformer block’s attention, and the rest of the model is frozen. Trainable params drop from 8 B (full FT) to ≈ 1 M; the resulting model still has the structure of attending to extra K/V every forward pass, which is mechanically much closer to “context tokens in the input.” This was our hypothesis going in.

We trained a prefix (16 virtual tokens, direct per-layer K/V, no projection encoder) on the same synthetic (Q, A) pairs as context-distillation, with learning rate 10^{-2} for ten epochs. Final training loss reached 0.001–0.020, comparable to full fine-tuning, and only ≈ 1 M parameters were updated.

Activation alignment is mixed. Figure 7 shows the three pairwise cosines per layer. For *knowledge-like* contexts (fewshot translation, factual injection), the prefix achieves cosine alignment with v^{ctx} that matches or exceeds full distillation ($\cos = 0.62$ for fewshot, vs 0.62 for distill; $\cos = 0.53$

Table 6: MMLU-Pro accuracy comparison including prefix-tuned model. Prefix tuning, despite using only ≈ 1 M trainable parameters, does *not* preserve general capability. For the haiku context in particular, prefix tuning collapses MMLU-Pro from 0.36 to 0.07 — worse than full distillation’s preserved 0.37.

Context	base	in-context	context-distilled	prefix	Δ prefix
haiku	0.360	0.295	0.365	0.070	− 0.290
pirate	0.360	0.325	0.125	0.145	−0.215
concise	0.360	0.380	0.300	0.085	− 0.275
fewshot_translate_fr	0.360	0.375	0.135	0.135	−0.225
factual	0.360	0.320	0.140	0.125	−0.235

for factual, vs 0.54). For *constraint-like* contexts (haiku, concise, pirate), the prefix’s alignment (0.28–0.56) lags behind full distillation (0.66–0.76).

This makes sense mechanistically: prefix tuning provides extra K/V tokens for attention to read, which is well-suited to “additional content to retrieve” (a fact, a translation example) but poorly suited to “a constraint on output form” (5-7-5 haiku, one-sentence answers, pirate vocabulary). The structure-constraint contexts seem to need the model’s output computation itself to change, which prefix tuning does not allow.

But MMLU-Pro accuracy is destroyed too. Table 6 reports MMLU-Pro accuracy with the prefix-tuned model added as a fourth variant. The result is striking: **prefix tuning destroys MMLU-Pro accuracy for every context, in some cases more severely than full distillation.** For haiku, the prefix-tuned model crashes from 0.36 to 0.07 — below the random baseline that one might expect from the floor of a 10-option MCQ.

This is the mechanism we missed. Even though the prefix preserves the structure of “extra K/V for attention to read,” the model has been *trained* to use the prefix to produce context-specific outputs. At inference time on out-of-distribution queries, the prefix is still active in every attention layer, and the model has lost the ability to ignore it. Selectivity is broken not by the K/V’s existence but by the training signal that conditioned the model on its presence.

Diagnosis: bake-in is the problem. Both full distillation and prefix tuning train the model to act context-conditioned given C ’s effect, with C no longer in the input. Once trained, the model cannot tell when C is supposed to apply — it always does. In-context preserves selectivity because C is in the input itself: the model’s attention to C ’s tokens naturally drops when the rest of the prompt makes C irrelevant. The lesson is that “bake the context into the weights/parameters” is not the right primitive for context compression. A correct solution would need either (a) a content-conditional gate on whether to apply the absorbed context, or (b) keeping the context’s tokens in the input but compressed (e.g., a learned summary token), so that attention retains its selectivity. Both are open directions for future work.

8 Related work

In-context learning as implicit gradient descent. von Oswald et al. [2023], Akyürek et al. [2023], and Garg et al. [2022] establish, in linear- or restricted-attention settings, that a forward pass over an in-context demonstration is mathematically equivalent to one step of gradient descent on a meta-objective. They characterise what in-context is doing internally, without ever performing the explicit gradient descent on the real transformer’s weights. Our work runs the empirical sibling experiment: context-distillation *actually* performs gradient descent on the model’s weights and we measure how close the resulting activation pattern is to the in-context one. The peak cosine of 0.42–0.80 supports the qualitative prediction on a real instruction-tuned transformer, but the magnitude gap (context-distilled overshooting in-context by a factor that grows with depth) shows the empirical equivalence is directional, not exact.

Knowledge editing. ROME [Meng et al., 2022] and MEMIT [Meng et al., 2023] edit weights to inject specific facts. Our factual-context experiment is the dual operation: rather than choosing the edit, we let SGD discover one and ask whether the result resembles the context that prompted it.

Soft prompts and prompt distillation. Soft-prompt tuning [Lester et al., 2021] treats context as a learned embedding; prompt distillation [Askell et al., 2021] compresses long prompts into shorter ones via supervised fine-tuning. The latter is methodologically close to our distillation step, but it is consumed as a training procedure, not as a mechanistic probe.

9 Discussion

Same direction, different amount. In-context and context-distilled produce *aligned but not identical* residual-stream perturbations. The cosine reaches 0.42–0.80 in late layers, but $\|v^{\text{dist}}\| > \|v^{\text{ctx}}\|$ at every layer with the gap widening with depth. The two interventions are related but not interchangeable.

Bake-in fails regardless of capacity. The MMLU-Pro and prefix-tuning experiments reveal a deeper structural issue: *any* fine-tune that compresses C into model parameters — whether 8 B parameters of full FT or 1 M parameters of prefix K/V — destroys general capability on unrelated tasks, often more severely than the smaller-parameter approach. The shared failure mode is the bake-in itself: the absorbed context is always active on every forward pass, with no mechanism for the model to recognise when it is irrelevant. In-context preserves capability because attention can choose to ignore C ’s tokens based on the rest of the input. Whatever the right primitive for “compressing context into model memory” is, it is not bake-in.

Implications for the in-context-learning-as-GD literature. The line of work from von Oswald et al. [2023] and follow-ups predicts that explicit GD on a suitable meta-loss should match the in-context forward pass. Our cosine of 0.42–0.80 is consistent with that prediction extended to a real 8B transformer with a real CE loss; the gap from 1.0 and the magnitude overshoot together suggest either that real CE-on-tokens is not exactly the meta-loss those papers identify, or that 32 nonlinear layers leave many descent directions equivalent for the same behavior — and SGD picks one that goes *further* along that direction than the in-context computation does.

Why magnitudes diverge. We do not have a complete account, but a plausible reading is that context-distillation is trained to reproduce specific outputs, so gradient pressure does not stop once the right direction is found — it continues until the loss is minimised. In-context, by contrast, is the result of a single forward pass that has no such optimisation pressure: it produces just enough perturbation to elicit the context-conditioned behavior. Two routes to similar outputs, with different stopping points.

10 Limitations

- Single model family (Llama-3.1-8B-Instruct). Whether the in-context vs context-distilled comparison transfers to other architectures is open.
- Single training-set size, learning rate, and step count per context. Whether the magnitude overshoot persists with longer training, more data, or different optimisers is open.
- The synthetic distillation set is generated by the same base model that is being fine-tuned, so context-distilled is closer to behavior cloning of self-outputs than to imitation of an external teacher; results may underestimate the difficulty of context internalisation in general.
- We do not perform any weight-space comparison. The paper claims pertain only to activations.
- Single run per (context) configuration. No error bars across seeds or training-recipe variations.
- Probe and validation sets are small (≈ 30 queries each); a downstream-task benchmark would tighten the behavioral claim.

References

- Ekin Akyürek, Dale Schuurmans, Jacob Andreas, Tengyu Ma, and Denny Zhou. What learning algorithm is in-context learning? investigations with linear models. *ICLR*, 2023.
- Amanda Askell, Yuntao Bai, Anna Chen, Dawn Drain, Deep Ganguli, Tom Henighan, Andy Jones, Nicholas Joseph, Ben Mann, Nova DasSarma, et al. A general language assistant as a laboratory for alignment. *arXiv preprint arXiv:2112.00861*, 2021.
- Abhimanyu Dubey et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Shivam Garg, Dimitris Tsipras, Percy Liang, and Gregory Valiant. What can transformers learn in-context? a case study of simple function classes. *NeurIPS*, 2022.
- Brian Lester, Rami Al-Rfou, and Noah Constant. The power of scale for parameter-efficient prompt tuning. *EMNLP*, 2021.
- Xiang Lisa Li and Percy Liang. Prefix-tuning: Optimizing continuous prompts for generation. *ACL*, 2021.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. *NeurIPS*, 2022.
- Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. Mass-editing memory in a transformer. *ICLR*, 2023.
- Johannes von Oswald, Eyvind Niklasson, Ettore Randazzo, João Sacramento, Alexander Mordvintsev, Andrey Zhmoginov, and Max Vladymyrov. Transformers learn in-context by gradient descent. *ICML*, 2023.
- Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, et al. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. *NeurIPS Datasets and Benchmarks*, 2024.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and Section 1 list four contributions (protocol, context library, alignment map, behavior-verification gate); these are exactly what Sections 2–4 deliver.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 10 enumerates four limitations: single model family (Llama-3.1-8B), single training-set size and learning rate per context, self-distillation framing, and ad-hoc probe set.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: This paper is empirical; it does not include theoretical results requiring proof.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 2 specifies the model id, the five context strings verbatim, training hyperparameters (lr, epochs, batch size, optimiser), the synthetic-dataset construction protocol, and the residual-stream capture protocol. The probe-query bank is included in the supplementary code.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: All code (synthetic-data generation, distillation, residual-stream capture, analysis) is included as supplementary material with a top-level README documenting the four-phase pipeline. The synthetic datasets are shipped as JSONL.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 2 reports optimiser (8-bit AdamW), learning rate, batch size, gradient accumulation, epochs, and the train/probe split per context.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: We run a single distillation per context. The alignment-curve points in Fig 1 are means over ~ 28 held-out probe queries; we report point estimates without error bars due to the sprint timeline. Across-seed variability would strengthen the claims but is left to follow-up work.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.

- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Each context’s full pipeline (Phase 1 generation, Phase 2 fine-tune, Phase 3 verify, Phase 4 capture) runs on a single NVIDIA RTX A6000 (48 GB) in ~15–25 minutes; the five contexts run in parallel for ~25 minutes wall-clock total. Total compute budget is approximately 16 GPU-hours.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn’t make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The work uses a publicly available pretrained model under its license, involves no human subjects, and produces no datasets at risk of misuse beyond what is already inherent in Llama-3.1-8B-Instruct.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 9 discusses implications for fine-tuning, knowledge editing, and unlearning. The bidirectional equivalence framing has dual-use implications — it could ease prompt recovery from fine-tuned weights — which we flag as a direction for future work and adversarial evaluation.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: We do not release a new pretrained foundation model. The fine-tuned checkpoints are derivatives of Llama-3.1-8B-Instruct under the Llama 3.1 Community License and inherit its safeguards.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: Llama-3.1-8B-Instruct [Dubey et al., 2024] is used under the Meta Llama 3.1 Community License. Software dependencies (PyTorch, Transformers, bitsandbytes) are credited and used under their respective open-source licenses.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset’s creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: We release: (i) the synthetic (C, Q, A) datasets per context as JSONL; (ii) the four-phase pipeline code with READMEs; (iii) the alignment-curve analysis utilities. Trained model checkpoints are available on request; their license is inherited from Llama-3.1.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: No crowdsourcing or human-subjects research.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No human-subjects research.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: An LLM (Llama-3.1-8B-Instruct) is the *subject* of study. It is used to generate the synthetic distillation datasets and is the model whose residual stream we probe. This usage is described throughout Sections 2–4.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.